



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



School of Information Science

COURSE PLAN

Academic Year 2026-27 ODD SEMESTER

School/Department of Students	School of Information Science
Name of the Program(s) of Students	Bachelor of Computer Application
PRC Approval Ref. No.	
Semester/Year	V
Course Code & Name	CSA3426 Front-End Development using JavaScript
Credit Structure (L-T-P-C)	1-0-4-3
Contact Hours	75 [15T+60L]
Course In-Charge (IC)	Mr Suhel Khan
Course Instructor(s)	Mr Suhel Khan
Course URL	https://presidencyuniversity.linways.com

1. COURSE PRE-REQUISITES:

CSA1002 Web Design and Development

2. COURSE DESCRIPTION:

This intermediate course enables students to perform front-end development using Javascript, with emphasis on employability skills. The course covers key technologies and architectures that enables the student to design and implement front-end. On successful completion of this course, the student shall be able to pursue a career in front end development. The students shall develop strong problem-solving skills as part of this course.

3. COURSE OBJECTIVES:

The objective of the course is to familiarize the learners with the concepts of Front-end Development using Javascripts and attain Employability Skills through Experiential Learning techniques.

4. COURSE OUTCOMES:

TABLE 1: COURSE OUTCOMES		
CO Number	Statement of CO	Blooms Cognitive Level
	<i>On successful completion of the course the students shall be able to</i>	
CO1	Design and develop static web pages using HTML5 elements and CSS3	[Apply]
CO2	Develop responsive web pages using CSS, JavaScript and bootstrap.	[Apply]
CO3	Demonstrate the concepts of Angular.js to develop a web front-end.	[Apply]
CO4	Illustrate the concepts of React.js to develop a web front-end.	[Apply]
CO5	Implement dynamic data handling by parsing JSON structures.	[Apply]

5. MAPPING OF COURSE OUTCOMES WITH PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

5.1 PROGRAM OUTCOMES:

On successful completion of the Program, the students will be able to:

PO 1: Application of Domain Knowledge: Apply the domain knowledge such as mathematics, science and software engineering fundamentals into the Computer Application related professions.

PO 2: Problem Solving & Analysis: Identify, Formulate, Analyse and Solve Complex Scenarios related to Computer Applications.

PO 3: Design/development of Activities: Conceive, Design and Develop various activities of Computer Applications.

PO 4: Conduct Investigations of Events: Carry out Investigation of an event and draw logical conclusions based on critical thinking and analytical reasoning.

PO 5: Modern Tool usage: Effectively apply relevant ICT Tools and digital tools to carry out Computer Application Attributes.

PO 6: Research: Identify suitable Research Methods and report the findings.

PO 7: Profession and Society: Apply the knowledge of the values and beliefs of multicultural society and a global perspective in the profession.

PO 8: Ethics: Identify ethical issues and embrace ethical values in conduct of Profession.

PO 9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Express thoughts and ideas effectively in writing and oral communication

PO 11: Project Management and Finance: Ability to work independently, identify appropriate resources required for a project, and manage a project through to completion.

PO 12: Life-long Learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of societal and technological change.

TABLE 2a: CO-PO Mapping												
CO. No	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	H	M	L	–	M	–	–	–	–	L	–	
CO2	H	M	L	–	M	–	–	–	–	L	L	
CO3	H	M	L	–	M	–	–	–	–	L	L	
CO4	H	M	L	–	M	–	–	–	–	L	M	
CO5	H	M	L	–	M	–	–	–	–	L	–	

5.2 PROGRAM SPECIFIC OUTCOMES:

On successful completion of the Program, the students will be able to:

PSO1	Disciplinary knowledge: Capable of demonstrating comprehensive knowledge and understanding of Computer Applications, Data Science and AI/ML techniques.
PSO2	Problem Solving: Identify, formulate and apply appropriate techniques in the areas related to Software development, big data, Network, Cloud computing technologies and related domains of varying complexities in real-time applications.
PSO3	Design/development of Applications: Conceive, Design and Develop Various Computer Applications and be able to apply data science and AI/ML techniques in specific domains such as healthcare, finance, agriculture, marketing, etc.,

TABLE 2b: CO-PSO Mapping			
CO Number	PSO1	PSO2	PSO3
CO1	H	H	M
CO2	H	H	M
CO3	H	H	M
CO4	H	H	M
CO5	H	H	M

6. COURSE CONTENT:

Module Number	Contents	Number of Sessions
1	Topics: Web development basics, Introduction to HTML5 structure, Semantic elements, Forms and inputs, Introduction to CSS3 styling, Selectors and properties, Box model, Flexbox and Grid, Introduction to JavaScript, Variables, Data types, Operators, Conditional statements, Loops, Functions.	4
2	Topics: JavaScript Events, DOM Manipulation, Form validation, Local and session storage, ES6 concepts (Arrow functions, Spread/Rest Operators, Destructuring), Introduction to Bootstrap, Grid system, Forms, Navigation bars, Buttons, Cards, Tables, Modal windows.	4
3	Topics: Understanding AJAX and asynchronous JavaScript, Fetch API vs. XMLHttpRequest, Handling JSON data, jQuery basics, Selectors, Effects (Hide, Show, Toggle, Fade, Slide), Event handling in jQuery, Animations, Creating a dynamic content loader with AJAX & jQuery.	4
4	Topics: Introduction to AngularJS, Directives, Controllers, Data binding, Routing, Creating Angular components, Fetching API data with Angular, Introduction to Django, Creating views and templates, Static files and media, Connecting Django with Angular for dynamic web applications.	3

REFERENCE MATERIALS:

TextBooks:

T1: Fender, Young, "Front-end Fundamentals", Leanpub, 2015

T2: Northwood, Chris, "The Front End Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer", APress, 2018

Reference Books:

R1: Flanagan D S, "Javascript: The Definitive Guide", 7th Edition, O'Reilly Media, 2020

R2: Alex Libby, Gaurav Gupta, and Asoj Talesra, "Responsive Web Design with HTML5 and CSS3 Essentials", Packt Publishing, 2016

Online / E-Resources

1. <https://developer.mozilla.org/en-US/>
2. <https://javascript.info/>
3. <https://getbootstrap.com/docs/5.0/getting-started/introduction/>
4. <https://docs.djangoproject.com/en/stable/>

7. DETAILED SCHEDULE OF INSTRUCTION

TABLE 3: LESSON PLAN				
Session No.	Topic	Sub-Topic	CO No.	Reference
1	Overview of the Course, Scopes and Opportunities, Detailed overview of modules	Course objectives, learning outcomes, career opportunities in front-end development, course roadmap	CO1	T1
Module 1				
2	Web development basics, HTML5 structure, Semantic elements, Forms and inputs	Client-server architecture, HTML document structure, semantic tags, form controls, input validation attributes	CO1	T1
3	CSS3 styling, Selectors, Box model, Flexbox and Grid	CSS syntax, selector types, box model properties, responsive layouts using Flexbox and CSS Grid	CO1	T1
4	Introduction to JavaScript, Variables, Data types, Operators	JavaScript syntax, variable declarations (var, let, const), primitive and non-primitive data types, arithmetic and logical operators	CO1	T2
5	Conditional statements, Loops, Functions	if-else statements, switch-case, for/while loops, function declaration, function expressions and scope	CO1	T2
Module 2				

6	JavaScript Events, DOM Manipulation, Form validation	Event handling, event listeners, DOM traversal and modification, client-side form validation techniques	CO2	T2
7	Local/session storage, ES6 concepts (Arrow functions, Spread/Rest, Destructuring)	Browser storage APIs, arrow functions, template literals, spread/rest operators, destructuring assignment	CO2	T2
8	Bootstrap – Grid system, Forms, Navigation bars	Bootstrap container system, responsive grid layouts, form components, navigation menus	CO2	T2
9	Bootstrap – Buttons, Cards, Tables, Modal windows	UI components, responsive tables, card layouts, modal dialogs, utility classes	CO2	T2
Module 3				
10	AJAX, asynchronous JavaScript, Fetch API vs. XMLHttpRequest	Synchronous vs asynchronous programming, AJAX workflow, promises, Fetch API implementation, XMLHttpRequest basics	CO5	R1
11	Handling JSON data, jQuery basics, Selectors	JSON parsing and stringification, jQuery syntax, DOM selection, method chaining	CO5	R1
12	jQuery Effects (Hide, Show, Toggle, Fade, Slide), Event handling, Animations	Visual effects, event binding, animation methods, callback functions	CO5	R2
13	Creating a dynamic content loader with AJAX & jQuery	AJAX requests, fetching remote data, updating DOM dynamically, error handling	CO5	Experiential Learning

Module 4				
14	Introduction to AngularJS, Directives, Controllers, Data binding, Routing, Angular components	Angular architecture, directives, controllers, two-way data binding, routing configuration, component-based development	CO3	T1
15	Introduction to Django, Views and templates, Static files, Connecting Django with Angular	Django project structure, URL routing, views, template rendering, static file management, frontend-backend integration	CO3	Online Resources

The main pedagogical methods in the course are as follows:

- Practical & Hands-on
- Project-based Learning
- Lecture Mode & PowerPoint Presentation
- Seminar by Students
- Video-based Learning
- Problem-based Learning Method

TABLE 4: SPECIAL DELIVERY METHOD

S. No	Session Number	Subtopic (as per Lesson Plan)	Pedagogical Method
1	3	Module 1: CSS3 Styling, Selectors, Box model, Flexbox and Grid	Self Learning
2	7	Module 2: ES6 Concepts – Arrow Functions, Spread/Rest Operators, Destructuring	Technology Enabled Learning
3	10	Module 3: AJAX and Asynchronous JavaScript, Fetch API vs. XMLHttpRequest	Participative Learning
4	14	Module 4: AngularJS – Directives, Controllers, Data Binding, Routing	Participative Learning
5	15	Module 4: Introduction to Django, Connecting Django with Angular	Technology Enabled Learning

S.No.	Lab Experiments	No of Sessions
1	Familiarization of HTML and CSS basics.	2
2	Create an HTML webpage showcasing biodata with CSS styling.	2
3	Design an interactive web page for a new restaurant using CSS3 features.	2
4	Create a simple web form to gather user information.	2
5	Practice basic JavaScript exercises, including creating a canvas drawing application.	2
6	Implement JavaScript exercises for form validation.	2
7	Create a student registration form using JavaScript.	2
8	Design an RSVP form using Bootstrap form controls.	2
9	Create a responsive image grid using Bootstrap 5.	2

10	Write a JavaScript program using AJAX to dynamically load content and implement jQuery effects like fading.	2
11	Create an AngularJS application module and controller in app.js.	4
12	Design an 'AngularJS Solar System Explorer' for planet data visualization.	4
13	Develop a simple Django app that displays a list of fruits and selected students for an event.	4
14	Develop a layout.html with a suitable header (navigation menu) and footer with copyright and developer information.	4
15	Inherit layout.html and create three additional pages: Contact Us, About Us, and Home page.	4
16	Implement AJAX and Fetch API for dynamic data loading from public APIs.	4
17	Create a basic React.js application using components, props, state, and hooks.	4
18	Develop multi-page React applications using React Router and API integration.	4
19	Create REST APIs using Express.js or FastAPI and perform CRUD operations.	4
20	Develop a mini full-stack web application integrating React frontend with Express.js/FastAPI backend.	4

8. ASSESSMENT SCHEDULE

TABLE 5: ASSESSMENT SCHEDULE						
Sl. No	Assessment Type	Coverage	CO Number(s)	Duration in Minutes	Marks	Weightage
1.	Continuous Assessment-I	Module-1 Exp 1 to 4	CO1	60	25	12.5%
2.	Midterm Exam	Module- 1&2 Exp 1 to 8	CO1 & CO2	90	50	25%
3.	Continuous Assessment -II	Module- 3 Exp 8 to 12	CO3 & CO4	60	25	12.5%
4.	Continuous Assessment - III	Module- 1, 2, & 3 Exp 12 to 16	CO4	60	25	12.4%
5.	Continuous Assessment - IV	Module- 4 Exp 17 to 20	CO5	60	25	12.5%
6.	End Term Examination	Module-1,2,3 &4 Exp 1 to 20	CO1, CO2, CO3, CO4, CO5	180	50	25%

9. COURSE CLEARANCE CRITERIA:

This is in accordance with the Academic Regulations of the University and the Program Regulations and Curriculum of the respective program.

10. SAMPLE QUESTIONS:

TABLE 6: SAMPLE QUESTIONS				
Sl. No	Question	Marks	CO Number	Blooms Cognitive Level
1	What are Semantic elements in HTML5 and why are they important for web development?	10	CO1	Remember
2	Explain how DOM Manipulation and Event handling work in JavaScript with an example.	10	CO2	Understand
3	Write a jQuery program that loads data from a server using AJAX and displays it on the page with a fade-in effect.	10	CO5	Apply
4	Create an AngularJS application with data binding and routing between two pages.	10	CO3	Apply

Sample Thought Provoking Questions to be asked to assess the Students' Preparedness to carry out the Task [For Laboratory Component]:

TABLE 7b: SAMPLE THOUGHT PROVOKING QUESTIONS(LAB)			
Sl No.	Question	Task No.	Course Outcome No.
1	What is the difference between inline, internal, and external CSS? When would you use each?	1	CO1
2	How does the Bootstrap grid system use breakpoints to create responsive layouts?	9	CO2
3	Write a JavaScript AJAX call to fetch JSON data from a public API and display the result dynamically on the page.	10	CO5
4	Create an AngularJS module with a controller that displays a list of students with their marks using ng-repeat.	11	CO3

11.MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

TABLE 7: SDG MAPPING			
S. No	Topic	SDG Number	Justification
1	HTML5, CSS3 – Static Web Pages	SDG9	Builds digital literacy and foundational web skills for quality education and innovation in industry (SDG 9).

12.CRITERIA FOR COURSE OUTCOME ATTAINMENT CALCULATION:

TABLE 8: Threshold and Target Set for Course Outcomes				
Sl. No	C.O. No.	Course Outcomes	Threshold in %	Target in %
1.	CO1	Design and develop static web pages using HTML5 elements and CSS3	60	60
2.	CO2	Develop responsive web pages using CSS, JavaScript and Bootstrap	50	50
3.	CO3	Demonstrate the concepts of Angular.js to develop a web front-end	55	55
4.	CO4	Illustrate the concepts of React.js to develop a web front-end	50	50
5.	CO5	Implement dynamic data handling by parsing JSON structures	50	50

13.SUMMARY:

Sl. No.	Activity	Start Date	End Date	Total Sessions
1	Program Integration & Overview of the Course	05-08-2026	05-08-2026	1
2	Module 01 & Exp 1 to 4	06-08-2026	29-08-2026	16
3	Continuous Assessment -I	09-09-2026	12-09-2026	2
4	Module 02 & Exp 5 to 7	01-09-2026	15-09-2026	17
5	Midterm Exam	05-10-2026	09-10-2026	-
7	Module 03 & Exp 7 to 10	12-09-2026	01-10-2026	16
8	Continuous Assessment -II (Problem Solving Assignment)	31-10-2026	03-11-2026	2
9	Module 04 – & Exp 11 to 15	03-10-2026	05-11-2026	17
10	Continuous Assessment -III	25-09-2026	25-09-2026	2
11	Continuous Assessment -IV (Project)	06-11-2026	15-11-2026	2
12	End-Term	07-12-2026	30-12-2026	-

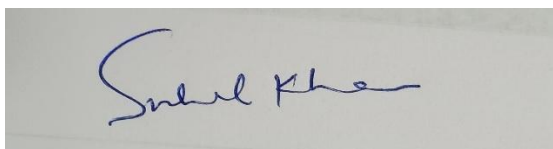
CONTACT TIMINGS IN THE CHAMBER FOR DISCUSSION

Students can meet the respective course instructor during the Chamber Consultation Hour to clarify doubts related to the course.

Name	Monday	Tuesday	Wednesday	Thursday	Friday
Mr Suhel Khan	3 to 4 pm	3 to 4 pm	3 to 4 pm	3 to 4 pm	3 to 4 pm

SPECIFIC GUIDELINES TO STUDENTS, IF ANY:

- Students are required to maintain minimum 75% of attendance
- Students are required to maintain classwork which will be evaluated at the end of every month
- Follow the instructions of course instructor in both class and lab.
- Avoid being absent in labs as it will affect the understanding of the experiments.
- On time submission of assignments.



Name and Signature of the course In-Charge

APPROVAL:

This course has been duly verified and approved by the Departmental Academic Committee (DAC).

Name and Signature of the Chairperson - DAC